VIRTUAL EMBRYO CHALLENGE

Challenge at a glance

Three prediction tasks, two team tracks, one final hidden-test phase.

T1

Temporal expression

single-cell RNA

future expression distribution

T2

Spatial-temporal

3D MERFISH

expression + 3D position

T3

Perturbation

3D MERFISH

knockout response + 3D

P2 now

Validation leaderboard

20 OCT

P3 opens: validation answers released

2 DEC

Final due

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All task splits

Training is public. Validation is scored in P2 and released at P3. Test ground truth stays hidden.

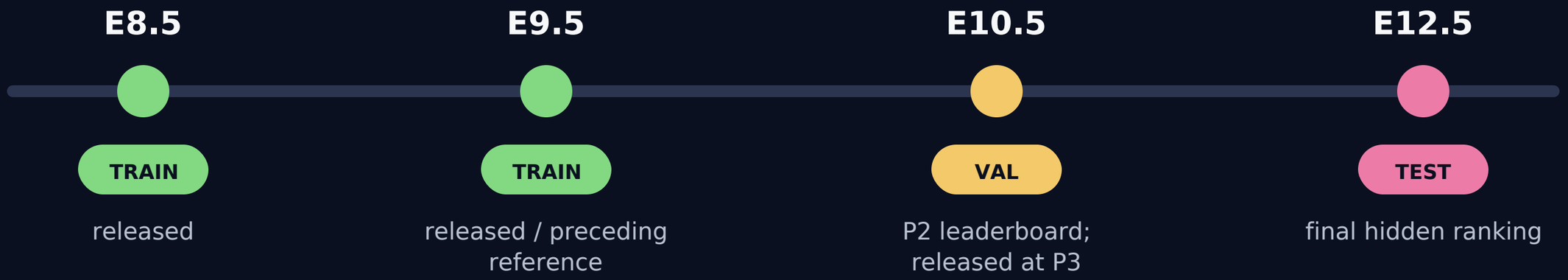
	TRAIN	VALIDATION	HIDDEN TEST
T1	E8.5 + E9.5	E10.5	E12.5
T2 heart	E8.25 / E8.75 / E9.5	E8.5 interp + E10.5 extrap	E12.5 extrap
T2 embryo	E6.75 / E7.25 / E8.0	E7.5 interp	E7.75 interp
T3	Mab21l2 KO + matched WT	Gata4 KO E8.75	beta-catenin KO E8.75

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Task 1 timeline

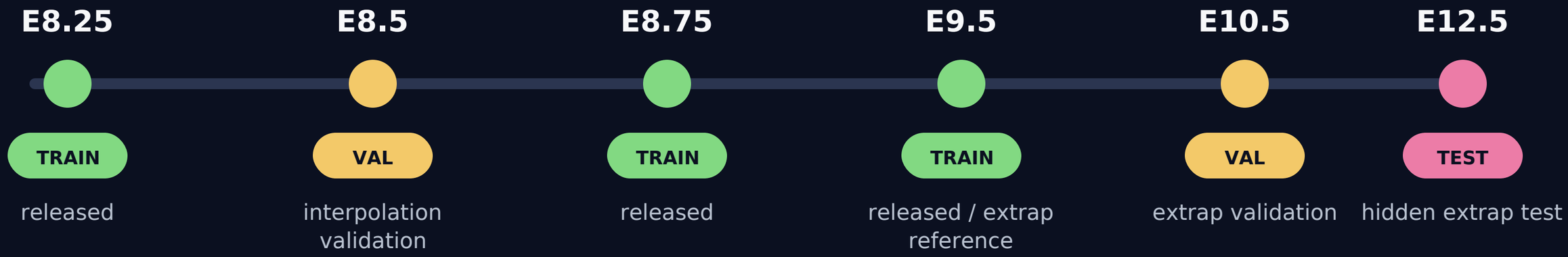
Temporal extrapolation: predict a future cell-expression distribution.



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Task 2 - heart

Interpolation and extrapolation live in one spatial-temporal setting.

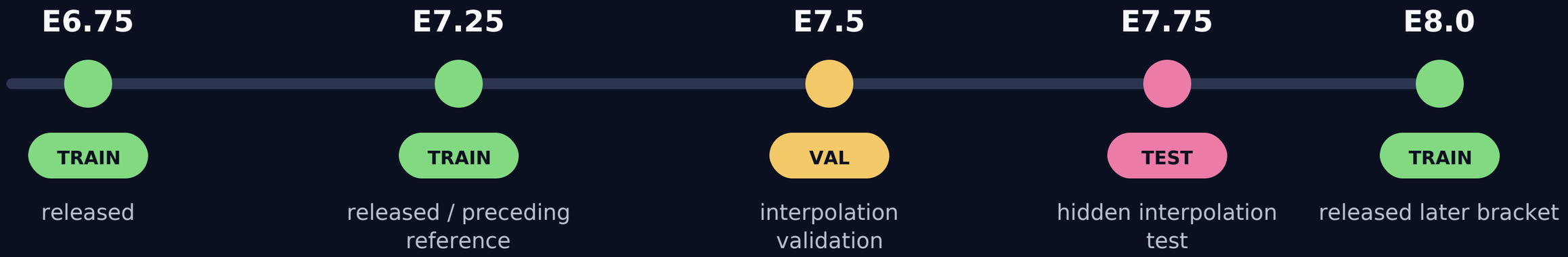


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Task 2 - whole embryo

Interpolation is the evaluated whole-embryo problem.



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Task 3 perturbation setup

Learn how a knockout changes an embryo, then transfer that response to a new gene.

REFERENCE

Matched WT

same stage



RELEASED PERTURBATION

Mab21l2 KO

E9.5



P2 VALIDATION

Gata4 KO

E8.75

P3 HIDDEN TEST

beta-catenin KO

E8.75

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Task 1 scoring map

Each metric asks a different biological question; the weighted skills become the leaderboard score.

25%	25%	30%	20%
DE recovery	**Direction**	**Population**	**Co-variation**

DE recovery

DES / de_score
higher is better

Direction

DCS / de_direction
higher is better

Population

MMD / mmd_u
lower is better

Co-variation

CSS / variogram
lower is better

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Task 2 scoring map

Four biological questions receive equal 25% weight; metric count does not determine importance.

25%	25%	25%	25%
Expression change	**Cell-state distribution**	**Shape + scale**	**Local organisation**

Expression change

DES + DCS
50/50 within group

Cell-state distribution

MMD + CSS
60/40 within group

Shape + scale

SDD + ODS + TSR
about one-third each

Local organisation

NFS / neighborhood_mmd
lower is better

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Task 3 scoring map

Task 3 scores the perturbation response. Absolute post-KO tissue shape is deliberately not a ranked question.

30%	25%	25%	20%
Gene recovery	**Direction**	**Magnitude**	**Cell states**

Gene recovery

DES / de_score
higher is better

Direction

DCS / de_direction
higher is better

Magnitude

severity_slope
zero is ideal

Cell states

MMD + CSS
60/40 within group

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Anatomy of a valid submission

What the scorer actually reads from your .h5ad file.

AnnData / .h5ad

.X — **cells x genes**

finite, non-negative, log-normalized

var_names — **exact released order**

whole transcriptome for T1; panel for spatial tasks

obsm["spatial_3D"] — **T2/T3 only**

Important scorer behavior

T1	no coordinates required
T2/T3	first 3 spatial columns are read
cells	predicted count need not equal target
celltype	submission labels are ignored
format	checks are unlimited in every phase

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P2 -> P3 transition

What changes on 20 October 2026 - and what does not.

P2 - NOW

Validation ground truth	hidden; score only
Leaderboard	validation
Scored attempts	8 per team/task/UTC day
Test answers	hidden

20 OCT

P3 - FINAL TEST

Validation ground truth	released for retraining
Leaderboard	hidden-test boards
Official attempts	2 per board / whole phase
Withdrawal	not allowed
Format checks	unlimited
Test ground truth	never distributed

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Agent Team evidence flow

A submission is held until at least two evidence kinds arrive, one of which must be the trajectory.

RUN

agent + model framework recorded

→

TRAJECTORY

what it tried, observed, did next

→

PROMPTS

every prompt given to the agent

→

HARNESS

code that ran the agent

Minimum to enter scoring queue

Trajectory + at least one of prompts or harness

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